



SIMATS ENGINEERING

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Course Code : CSA0509

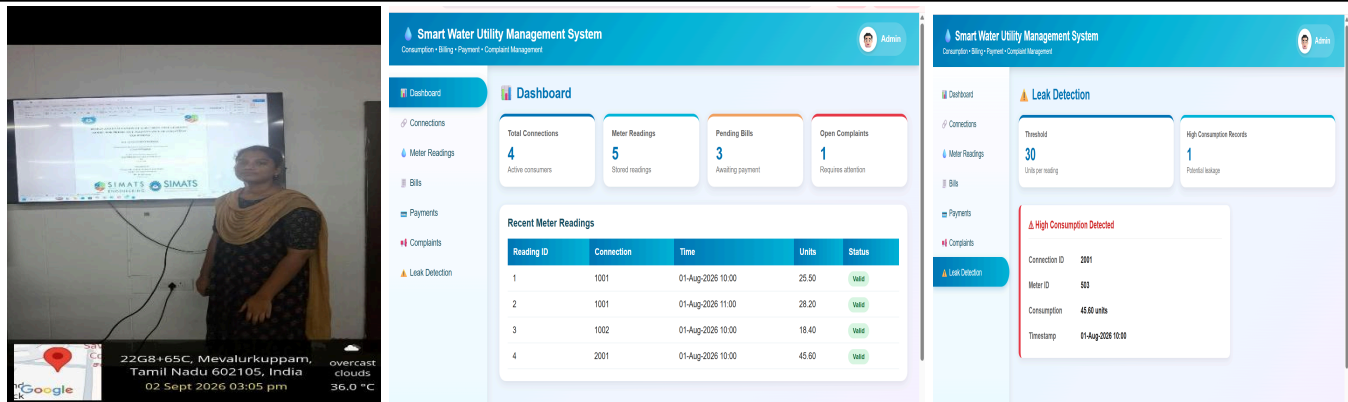
Slot: A

Course Name: Database Management Systems

Course Faculty: Dr. R. Kesavan

Project Title: Smart Water Utility Consumption, Billing, And Complaint Management System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

In this assignment, I worked on understanding the requirements of the Smart Water Utility Consumption, Billing, and Complaint Management System. The database structure was designed to manage connections, meter readings, bills, payments, and complaints. I created and populated the required tables using MySQL. Suitable file organization, indexing, and hashing techniques were selected to improve data storage and retrieval. SQL queries were developed for consumption history, leak detection, billing, payments, and complaint management. Query optimization was performed using indexes and EXPLAIN for analyzing execution plans. Transaction management was implemented using ACID concepts, COMMIT, ROLLBACK, and SAVEPOINT. Concurrency-control methods were considered to avoid duplicate bills and inconsistent payment updates. Testing was carried out using different validation cases, followed by debugging of SQL and implementation errors. The required diagrams, screenshots, results, and documentation were also prepared. The database design and query outputs were reviewed to ensure accurate results. Performance and data-retrieval efficiency were considered during implementation. The final work was compiled and organized for submission and presentation. Overall, the contribution focused on developing and validating an efficient, reliable, and consistent database solution.

Student Signature

Guide Signature